

Computer Modelling - Unit 3

Task 4.2

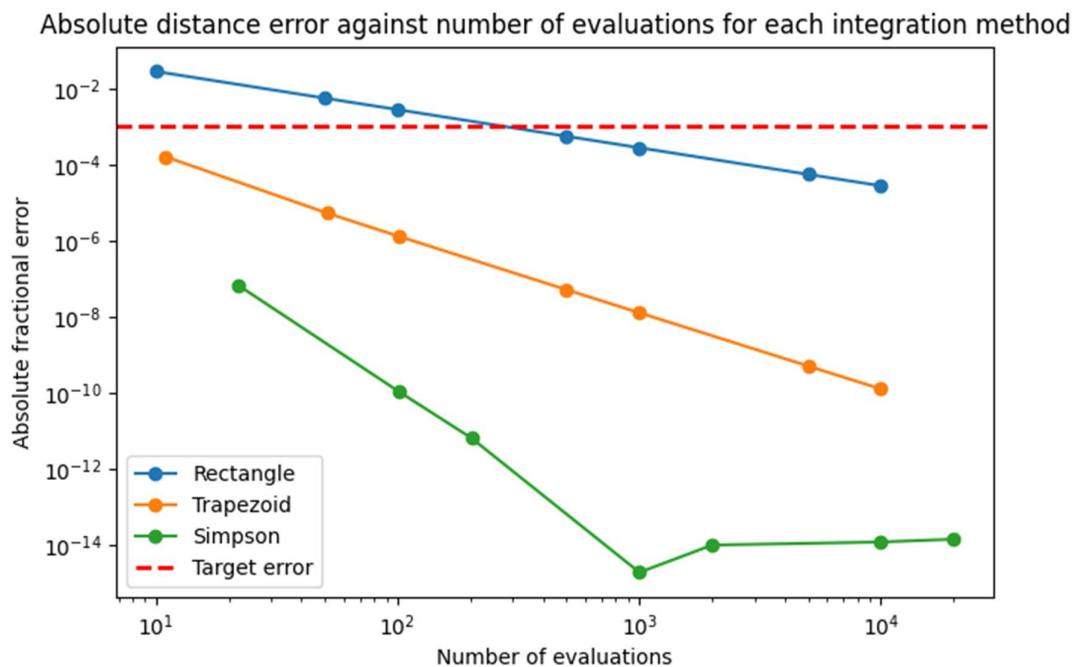
Distance using rectangle at 10000 steps: 3212.1457296019194

Distance using trapezoid at 10000 steps: 3212.0557839189014

Distance using Simpson's at 10000 steps: 3212.055783501712

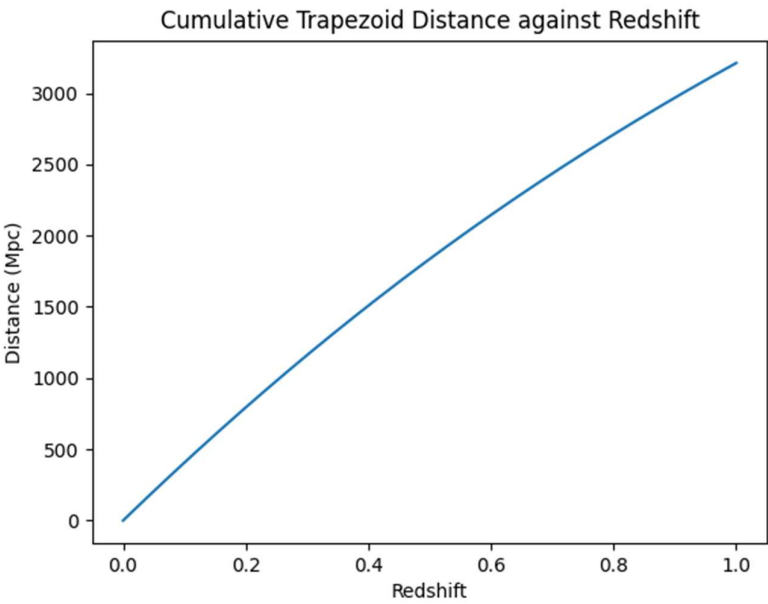
Number of steps	Rectangle Rule	Trapezoid Rule	Simpson Rule
100	3221.0545231663673	3212.0600395537654	3212.0557834801994
1000	3212.9552820453805	3212.055825297954	3212.055783501698
10000	3212.1457296019194	3212.0557839189014	3212.055783501712
100000	3212.064778074204	3212.055783505873	3212.055783501665

Target Accuracy: Distance to Andromeda galaxy can be measured with a 4% uncertainty at a distance of about 0.8Mpc [Distance to the Andromeda galaxy from eclipsing binaries, F. Vilardell et al, Astronomy&Astrophysics, 21/01/10]. It is also worth noting that currently the Hubble constant H_0 is known to about 1% uncertainty [The Hubble Tension, V.Poulin, CernCourier, <https://cerncourier.com/a/the-hubble-tension/#:~:text=One%20of%20the%20main%20predictions,The%20Hubble%20tension%E2%80%9D%20figure>). (accessed14/11/25)]. Considering this and the fact the distance to the examined galaxy is about 3200Mpc, 3 orders magnitude more than to Andromeda, I would estimate a target accuracy of 10^{-3} .



Recommended method: Using Simpson with 10^3 steps will give the lowest error for least evaluations, resulting in the most efficient calculation for the most accurate value.

Task 4.3



Task 4.4

